Resampling – Problem Set 2

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Problem 1

Suppose $\hat{\theta}$ is a linear statistic of the form

$$\hat{\theta} = \mu + \frac{1}{n}\sum_{i=1}^{n} \alpha(x_i).$$

Suppose we know the value of $\hat{\theta}$ for each jackknife dataset $x_{(-i)}$. Then it is given by b_i, for $i = 1, 2, \ldots, n$.

(a) Take $\alpha_i = \alpha(x_i)$ and solve the set of linear equations

$$b_i = \mu + \frac{1}{n-1}\sum_{j\neq i} \alpha_j, \qquad i = 1, 2, \ldots, n.$$

(b) Hence deduce the value of $\hat{\theta}^*$ for an arbitrary bootstrap sample $x_1^*, x_2^*, \ldots, x_n^*$.

(c) Calculate the estimates of the standard errors using jackknife and bootstrap. Show that

$$\mathrm{se}_{\text{jack}} = \sqrt{\frac{n-1}{n}}\, s^*_{\text{jack}}.$$

Problem 2

Suppose $\hat{\theta}$ is a quadratic statistic, i.e.

$$\hat{\theta} = \mu + \frac{1}{n}\sum_{i} \alpha_i(x_i) + \frac{1}{n^2}\sum_{i,j} \beta_{ij},$$

where $\alpha_i = \alpha(x_i)$ and $\beta_{ij} = \beta(x_i, x_j)$. Define

$$\hat{\theta}^*(w) = \mu + \sum_{i} w_i \alpha_i + \sum_{i,j} w_i w_j \beta_{ij}.$$

(i) Show that bootstrap replicates can be written as $\hat{\theta}^*(w)$ where

$$w \sim \frac{1}{n}\,\mathrm{Multinomial}\big(n, (1, 1, \ldots, 1)\big).$$

(ii) Show that jackknife replicates can be written as $\hat{\theta}^*(w)$ where $w = F_{n-1,i}$ with probability $\frac{1}{n}$, and $F_{n-1,i}$ is the empirical distribution function with X_i removed.

(iii) Find $E(w_i)$ and $E(w_i w_j)$ for both of the above cases.

(iv) Hence show that

$$\mathrm{bias}_{\text{boot}} = \frac{n-1}{n}\,\mathrm{bias}_{\text{jack}}.$$

Delta Method Note

The Delta Method is based on the Taylor series expansion of a smooth function $g(\cdot)$ around θ:

$$g(\hat{\theta}) = g(\theta) + g'(\theta)(\hat{\theta} - \theta) + \frac{1}{2}g''(\theta)(\hat{\theta} - \theta)^2 + R_2(\hat{\theta} - \theta)$$

where R_2 is a remainder term that vanishes quickly as $\hat{\theta} \to \theta$.

First-Order Delta Method

If $g'(\theta) \neq 0$, the linear term dominates. We can rearrange the expansion as:

$$\sqrt{n}(g(\hat{\theta}) - g(\theta)) \approx g'(\theta)\sqrt{n}(\hat{\theta} - \theta)$$

Since $\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, \sigma^2)$, by Slutsky's Theorem:

$$\sqrt{n}(g(\hat{\theta}) - g(\theta)) \xrightarrow{d} N(0, [g'(\theta)]^2\sigma^2)$$

Second-Order Delta Method

However, when $g'(\theta) = 0$, the first-order term vanishes. The behavior of the statistic is then governed by the next non-zero term in the Taylor expansion:

$$g(\hat{\theta}) - g(\theta) \approx \frac{1}{2}g''(\theta)(\hat{\theta} - \theta)^2$$

Multiplying by n, we get:

$$n(g(\hat{\theta}) - g(\theta)) \approx \frac{1}{2}g''(\theta)\left[\sqrt{n}(\hat{\theta} - \theta)\right]^2$$

Defining $Z \sim N(0, \sigma^2)$ as the limit of $\sqrt{n}(\hat{\theta} - \theta)$, we see that the distribution converges to a scaled Chi-squared distribution:

$$n(g(\hat{\theta}) - g(\theta)) \xrightarrow{d} \frac{1}{2}g''(\theta)Z^2$$

Problem 3

Suppose $X_i \sim N(\mu, 1)$ and $\hat{\theta} = \mu^2$. So $\hat{\theta} = \bar{X}^2$.

(i) Show that for $\mu = 0$, $n\bar{X}^2 \sim \chi^2_1$.

(ii) Show that for $\mu = 0$, the standard n-out-of-n bootstrap does not produce the correct limiting distribution.

(iii) Describe how the m-out-of-n bootstrap will fix this issue.

Problem 4

Suppose X_i has mean μ and variance 1. Let $\theta = \sqrt{\mu}$ and $\hat{\theta} = \sqrt{\bar{X}}$.

(i Using delta method, find out the limiting distribution of $\hat{\theta}$ and $\hat{\theta}^*$ when $\mu \neq 0$.

(ii) Show that for $\mu = 0$, the bootstrap distribution becomes inconsistent and its variance goes to infinity.

(iii) Show how the m-out-of-n bootstrap fixes this problem.